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Exp no:1

Setting up Hyperledger Fabric Development Environment

Aim

To install and configure the Hyperledger Fabric development environment on a Linux system using Docker, Go, and Fabric binaries, create a blockchain test network, and establish a communication channel between peer organizations

By the end of the experiment:

- Hyperledger Fabric installed
- Docker running
- Fabric binaries downloaded
- A blockchain network running
- A communication channel (mychannel) created between two organizations

Procedure:

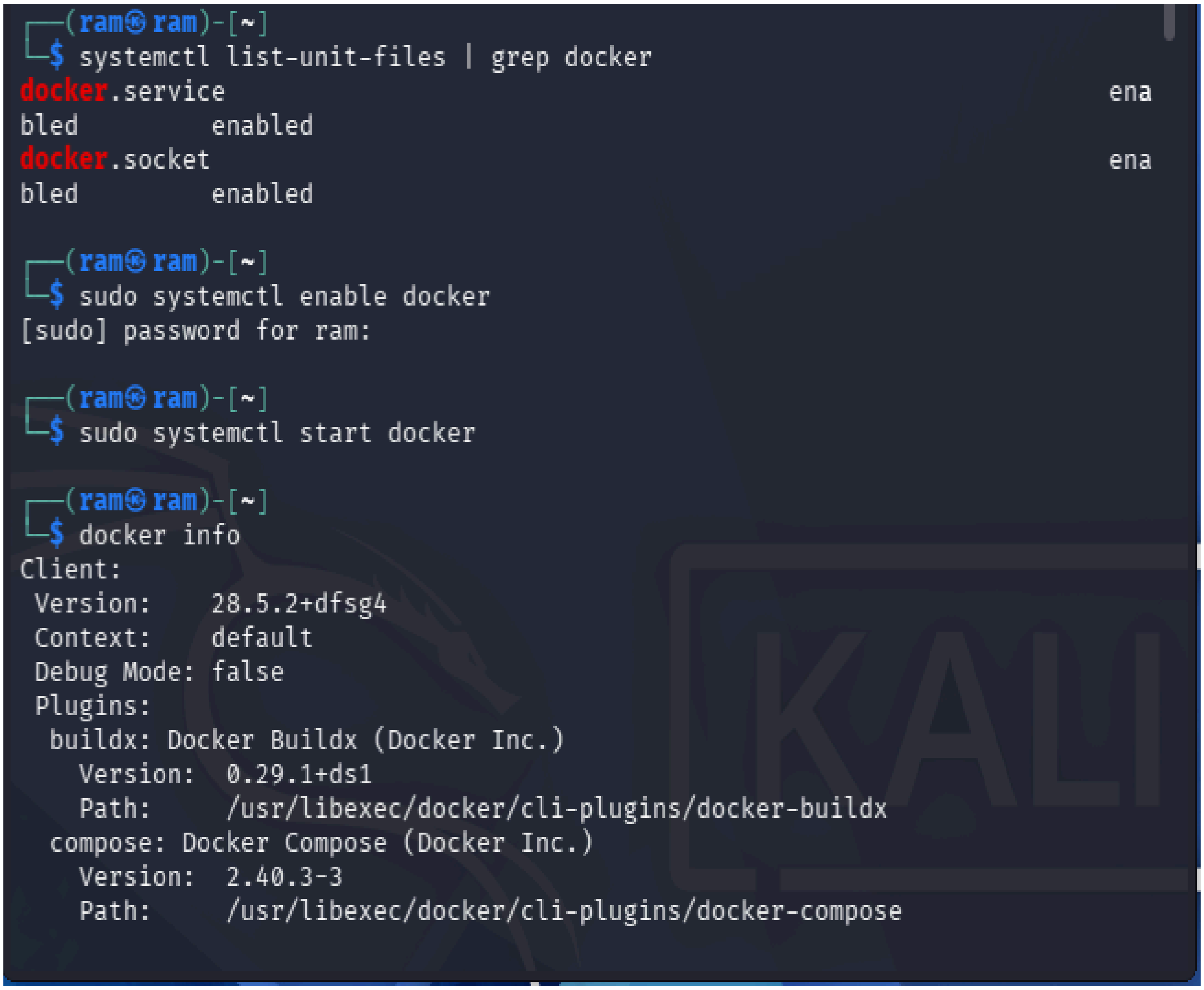
1. **sudo apt install -y git curl jq docker-compose**

this command uses administrative privileges to automatically download and install four developer tools—Git, Curl, JQ, and Docker Compose

2. **Verify the installation**

```
└─$ git --version
git version 2.53.0

┌──(ram㉿ram)-[~/fabric-dev/fabric-samples/test-network]
└─$ curl --version
curl 8.20.0 (x86_64-pc-linux-gnu) libcurl/8.20.0 OpenSSL/3.6.2 zlib/1.3.2 bro
tli/1.2.0 zstd/1.5.7 libidn2/2.3.8 libpsl/0.22.0 libssh2/1.11.1 nghttp2/1.69.
0 ngtcp2/1.22.1 nghttp3/1.15.0 mit-krb5/1.22.1 OpenLDAP/2.6.10
Release-Date: 2026-04-29, security patched: 8.20.0-5
Protocols: dict file ftp ftps gopher gophers http https imap imaps ipfs ipns
ldap ldaps mqtt mqtts pop3 pop3s rtsp scp sftp smtp smtps telnet tftp ws wss
Features: alt-svc AsynchDNS brotli GSS-API HSTS HTTP2 HTTP3 HTTPS-proxy IDN I
Pv6 Kerberos Largefile libz PSL SPNEGO SSL threadsafe TLS-SRP UnixSockets zst
d

┌──(ram㉿ram)-[~/fabric-dev/fabric-samples/test-network]
└─$ docker-compose --version
Docker Compose version 2.40.3-3

┌──(ram㉿ram)-[~/fabric-dev/fabric-samples/test-network]
└─$ ^C

┌──(ram㉿ram)-[~/fabric-dev/fabric-samples/test-network]
└─$ docker --version
Docker version 28.5.2+dfsg4, build 9cc6dea35e9a963f281434761c656fba4ac43aed
```

3. **Start and enable the docker:**



```
┌──(ram㉿ram)-[~]
└─$ systemctl list-unit-files | grep docker
docker.service                                                              ena
bled          enabled
docker.socket                                                               ena
bled          enabled

┌──(ram㉿ram)-[~]
└─$ sudo systemctl enable docker
[sudo] password for ram:

┌──(ram㉿ram)-[~]
└─$ sudo systemctl start docker

┌──(ram㉿ram)-[~]
└─$ docker info
Client:
 Version:    28.5.2+dfsg4
 Context:    default
 Debug Mode: false
 Plugins:
  buildx: Docker Buildx (Docker Inc.)
    Version:  0.29.1+ds1
    Path:     /usr/libexec/docker/cli-plugins/docker-buildx
  compose: Docker Compose (Docker Inc.)
    Version:  2.40.3-3
    Path:     /usr/libexec/docker/cli-plugins/docker-compose
```

4. **Add yourself to Docker group**

sudo usermod -aG docker $USER

5. **Refresh groups**

newgrp docker

6. **Test Docker**

docker run hello-world

```
┌──(ram㉿ram)-[~/fabric-dev/fabric-samples/test-network]
└─$ docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent i
t
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/

┌──(ram㉿ram)-[~/fabric-dev/fabric-samples/test-network]
```

7. **Check Go**

sudo apt install golang-go
go version

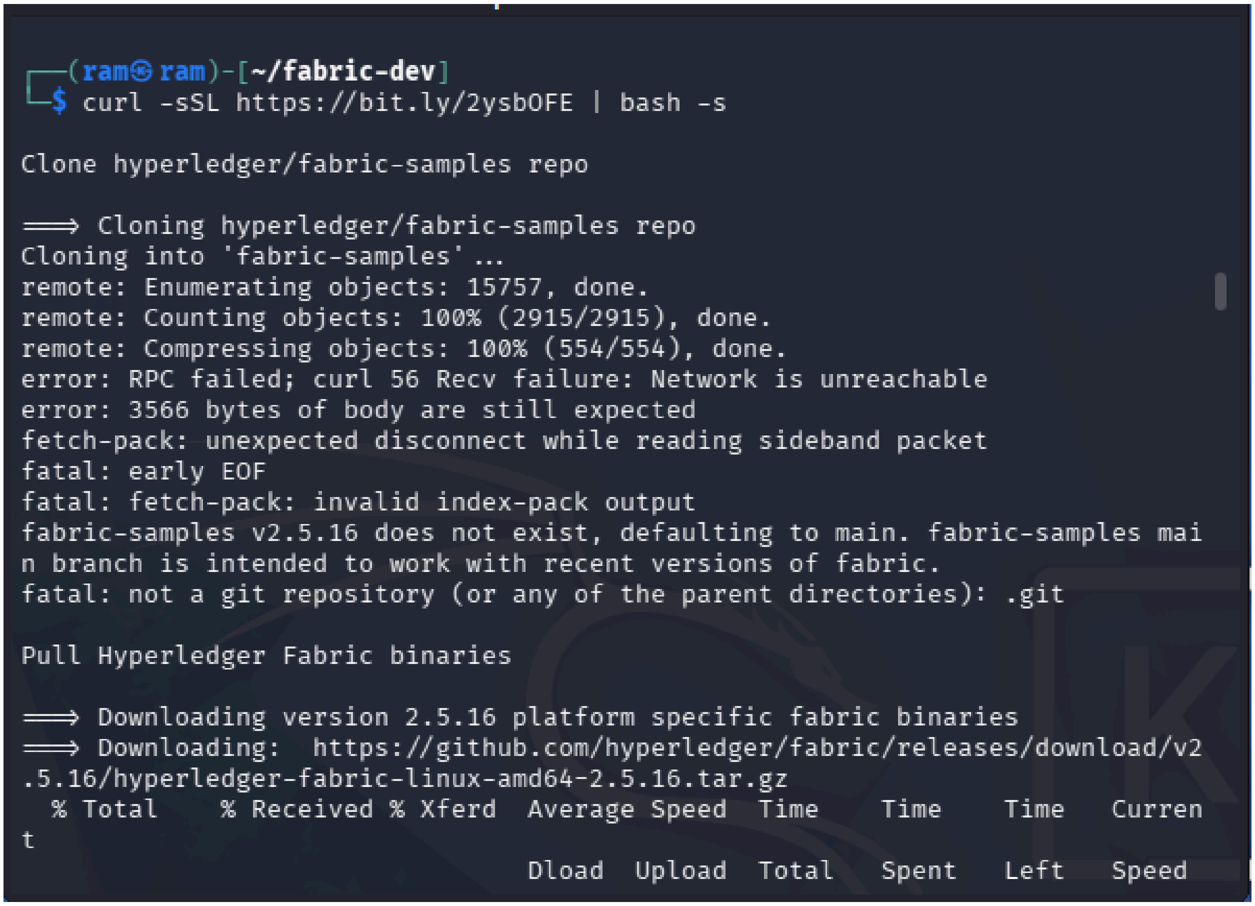
```
No VM guests are running outdated hypervisor (qemu) binaries on this host.

┌──(ram㉿ram)-[~]
└─$ go version
go version go1.26.5 linux/amd64

┌──(ram㉿ram)-[~]
└─$ mkdir -p ~/fabric-dev
```

8. **Download Hyperledger Fabric**

curl -sSL https://bit.ly/2ysbOFE | bash -s

This downloads:

Fabric binaries
Docker images
Sample applications



```
┌──(ram㉿ram)-[~/fabric-dev]
└─$ curl -sSL https://bit.ly/2ysbOFE | bash -s

Clone hyperledger/fabric-samples repo

===> Cloning hyperledger/fabric-samples repo
Cloning into 'fabric-samples'...
remote: Enumerating objects: 15757, done.
remote: Counting objects: 100% (2915/2915), done.
remote: Compressing objects: 100% (554/554), done.
error: RPC failed; curl 56 Recv failure: Network is unreachable
error: 3566 bytes of body are still expected
fetch-pack: unexpected disconnect while reading sideband packet
fatal: early EOF
fatal: fetch-pack: invalid index-pack output
fabric-samples v2.5.16 does not exist, defaulting to main. fabric-samples mai
n branch is intended to work with recent versions of fabric.
fatal: not a git repository (or any of the parent directories): .git

Pull Hyperledger Fabric binaries

===> Downloading version 2.5.16 platform specific fabric binaries
===> Downloading:  https://github.com/hyperledger/fabric/releases/download/v2
.5.16/hyperledger-fabric-linux-amd64-2.5.16.tar.gz
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Curren
t
                                 Dload  Upload   Total   Spent    Left  Speed
```

9. **Enter the Sample Folder**

cd fabric-samples

10. **Configure PATH**

echo 'export PATH=$PATH:$HOME/fabric-dev/fabric-samples/bin' >> ~/.bashrc
source ~/.bashrc

11. **Verify Installation**

Check Peer:
peer version

Check Orderer:
orderer version

```
┌──(ram㉿ram)-[~/fabric-dev/fabric-samples]
└─$ export PATH=$PATH:$HOME/fabric-dev/fabric-samples/bin

┌──(ram㉿ram)-[~/fabric-dev/fabric-samples]
└─$ peer version
peer:
 Version: v2.5.16
 Commit SHA: f871cf9
 Go version: go1.26.4
 OS/Arch: linux/amd64
 Chaincode:
  Base Docker Label: org.hyperledger.fabric
  Docker Namespace: hyperledger


┌──(ram㉿ram)-[~/fabric-dev/fabric-samples]
└─$ orderer version
orderer:
 Version: v2.5.16
 Commit SHA: f871cf9
 Go version: go1.26.4
 OS/Arch: linux/amd64
```

12. Start the Blockchain Network

Go into the test network
cd ~/fabric-dev/fabric-samples/test-network

13. Start it

./network.sh up

```
┌──(ram㉿ram)-[~/fabric-dev/fabric-samples]
└─$ cd test-network

┌──(ram㉿ram)-[~/fabric-dev/fabric-samples/test-network]
└─$ ./network.sh up
Using docker and docker-compose
Starting nodes with CLI timeout of '5' tries and CLI delay of '3' seconds and
 using database 'leveldb' with crypto from 'cryptogen'
LOCAL_VERSION=v2.5.16
DOCKER_IMAGE_VERSION=v2.5.16
/home/ram/fabric-dev/fabric-samples/test-network/../bin/cryptogen
Generating certificates using cryptogen tool
Creating Org1 Identities
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-org1.ya
ml --output=organizations
org1.example.com
+ res=0
Creating Org2 Identities
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-org2.ya
ml --output=organizations
org2.example.com
+ res=0
Creating Orderer Org Identities
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-orderer
.yaml --output=organizations
+ res=0
Generating CCP files for Org1 and Org2
[+] Running 7/7
 ✔ Network fabric_test                     Crea...                    0.0s
 ✔ Volume compose_peer0.org1.example.com   Created                    0.0s
 ✔ Volume compose_peer0.org2.example.com   Created                    0.0s
 ✔ Volume compose_orderer.example.com      Created                    0.0s
 ✔ Container peer0.org2.example.com        Started                    0.2s
 ✔ Container orderer.example.com           Started                    0.2s
 ✔ Container peer0.org1.example.com        Started                    0.1s
CONTAINER ID   IMAGE                               COMMAND             CREATE
D              STATUS                           PORTS


                             NAMES
f7f27b14a5a9   hyperledger/fabric-orderer:latest   "orderer"           Less t
han a second ago   Up Less than a second       0.0.0.0:7050->7050/tcp, [::]:7
050->7050/tcp, 0.0.0.0:7053->7053/tcp, [::]:7053->7053/tcp, 0.0.0.0:9443->944
3/tcp, [::]:9443->9443/tcp   orderer.example.com
a8a53b3de1fa   hyperledger/fabric-peer:latest      "peer node start"   Less t
han a second ago   Up Less than a second       0.0.0.0:9051->9051/tcp, [::]:9
051->9051/tcp, 7051/tcp, 0.0.0.0:9445->9445/tcp, [::]:9445->9445/tcp
                             peer0.org2.example.com
5d5fabee5bea   hyperledger/fabric-peer:latest      "peer node start"   Less t
han a second ago   Up Less than a second       0.0.0.0:7051->7051/tcp, [::]:7
051->7051/tcp, 0.0.0.0:9444->9444/tcp, [::]:9444->9444/tcp
                             peer0.org1.example.com
2492821887e5   hello-world                         "/hello"            57 min
utes ago           Exited (0) 57 minutes ago

                             fervent_dubinsky
```

14. **Create Channel**

./network.sh createChannel -c mychannel

This creates a communication channel named mychannel.

It also:
creates the channel
joins both organizations
configures anchor peers

```
┌──(ram㉿ram)-[~/fabric-dev/fabric-samples/test-network]
└─$ ./network.sh createChannel -c mychannel
Using docker and docker-compose
Creating channel 'mychannel'.
If network is not up, starting nodes with CLI timeout of '5' tries and CLI de
lay of '3' seconds and using database 'leveldb
Bringing up network
LOCAL_VERSION=v2.5.16
DOCKER_IMAGE_VERSION=v2.5.16
[+] Running 3/3
 ✔ Container orderer.example.com     Running                              0.0s
 ✔ Container peer0.org1.example.com  Running                              0.0s
 ✔ Container peer0.org2.example.com  Running                              0.0s
CONTAINER ID   IMAGE                               COMMAND              CREATE
D       STATUS                       PORTS

                     NAMES
f7f27b14a5a9   hyperledger/fabric-orderer:latest   "orderer"            21 sec
onds ago   Up 21 seconds                0.0.0.0:7050→7050/tcp, [::]:7050→705
0/tcp, 0.0.0.0:7053→7053/tcp, [::]:7053→7053/tcp, 0.0.0.0:9443→9443/tcp, [
::]:9443→9443/tcp   orderer.example.com
a8a53b3de1fa   hyperledger/fabric-peer:latest      "peer node start"    21 sec
onds ago   Up 21 seconds                0.0.0.0:9051→9051/tcp, [::]:9051→905
1/tcp, 7051/tcp, 0.0.0.0:9445→9445/tcp, [::]:9445→9445/tcp
                     peer0.org2.example.com
5d5fabee5bea   hyperledger/fabric-peer:latest      "peer node start"    21 sec
onds ago   Up 21 seconds                0.0.0.0:7051→7051/tcp, [::]:7051→705
1/tcp, 0.0.0.0:9444→9444/tcp, [::]:9444→9444/tcp
                     peer0.org1.example.com
2492821887e5   hello-world                         "/hello"             58 min
utes ago   Exited (0) 58 minutes ago

                     fervent_dubinsky
Using docker and docker-compose
Generating channel genesis block 'mychannel.block'
Using organization 1
/home/ram/fabric-dev/fabric-samples/test-network/../bin/configtxgen
+ '[' 0 -eq 1 ']'
+ configtxgen -profile ChannelUsingRaft -outputBlock ./channel-artifacts/mych
annel.block -channelID mychannel
2026-07-16 05:01:32.908 IST 0001 INFO [common.tools.configtxgen] main → Load
ing configuration
2026-07-16 05:01:32.911 IST 0002 INFO [common.tools.configtxgen.localconfig]
completeInitialization → orderer type: etcdraft
2026-07-16 05:01:32.911 IST 0003 INFO [common.tools.configtxgen.localconfig]
completeInitialization → Orderer.EtcdRaft.Options unset, setting to tick_int
erval:"500ms" election_tick:10 heartbeat_tick:1 max_inflight_blocks:5 snapsho
t_interval_size:16777216
2026-07-16 05:01:32.911 IST 0004 INFO [common.tools.configtxgen.localconfig]
Load → Loaded configuration: /home/ram/fabric-dev/fabric-samples/test-networ
k/configtx/configtx.yaml
2026-07-16 05:01:32.912 IST 0005 INFO [common.tools.configtxgen] doOutputBloc
k → Generating genesis block
2026-07-16 05:01:32.912 IST 0006 INFO [common.tools.configtxgen] doOutputBloc
k → Creating application channel genesis block
2026-07-16 05:01:32.912 IST 0007 INFO [common.tools.configtxgen] doOutputBloc
k → Writing genesis block
+ res=0
Creating channel mychannel
Adding orderers
```

```
┌──(ram㉿ram)-[~/fabric-dev/fabric-samples/test-network]
└─$ ls channel-artifacts
config_block.json              Org1MSPanchors.tx
config_block.pb                Org1MSPconfig.json
config_update_in_envelope.json Org1MSPmodified_config.json
config_update.json             Org2MSPanchors.tx
config_update.pb               Org2MSPconfig.json
modified_config.pb             Org2MSPmodified_config.json
mychannel.block                original_config.pb

┌──(ram㉿ram)-[~/fabric-dev/fabric-samples/test-network]
└─$ mychannel.block
bash: mychannel.block: command not found

┌──(ram㉿ram)-[~/fabric-dev/fabric-samples/test-network]
└─$ ls organizations/peerOrganizations/org1.example.com
ca  connection-org1.json  connection-org1.yaml  msp  peers  tlsca  users

┌──(ram㉿ram)-[~/fabric-dev/fabric-samples/test-network]
└─$ docker ps
CONTAINER ID   IMAGE                               COMMAND             CREATE
D          STATUS          PORTS


       NAMES
f7f27b14a5a9   hyperledger/fabric-orderer:latest   "orderer"           4 minu
tes ago   Up 4 minutes   0.0.0.0:7050->7050/tcp, [::]:7050->7050/tcp, 0.0.0.0
:7053->7053/tcp, [::]:7053->7053/tcp, 0.0.0.0:9443->9443/tcp, [::]:9443->9443
/tcp   orderer.example.com
a8a53b3de1fa   hyperledger/fabric-peer:latest      "peer node start"   4 minu
tes ago   Up 4 minutes   0.0.0.0:9051->9051/tcp, [::]:9051->9051/tcp, 7051/tc
p, 0.0.0.0:9445->9445/tcp, [::]:9445->9445/tcp
       peer0.org2.example.com
5d5fabee5bea   hyperledger/fabric-peer:latest      "peer node start"   4 minu
tes ago   Up 4 minutes   0.0.0.0:7051->7051/tcp, [::]:7051->7051/tcp, 0.0.0.0
:9444->9444/tcp, [::]:9444->9444/tcp
       peer0.org1.example.com

┌──(ram㉿ram)-[~/fabric-dev/fabric-samples/test-network]
└─$ ls channel-artifacts
config_block.json              Org1MSPanchors.tx
config_block.pb                Org1MSPconfig.json
config_update_in_envelope.json Org1MSPmodified_config.json
config_update.json             Org2MSPanchors.tx
config_update.pb               Org2MSPconfig.json
modified_config.pb             Org2MSPmodified_config.json
mychannel.block                original_config.pb
```